Chapter 15, Problem 26P

Problem

Determine the initial and final values of $f(t)$, if they exist, given that:

(a) $F(s) = \dfrac{5s^2 + 3}{s^3 + 4s^2 + 6}$

(b) $F(s) = \dfrac{s^2 - 2s + 1}{4(s-2)(s^2 + 2s + 4)}$

Step-by-step solution

Step 1 of 6

(a)

The function $F(s)$ is,

$$F(s) = \frac{5s^2 + 3}{s^3 + 4s^2 + 6}$$

The initial value of any function is the value obtained at time $t = 0$ in time domain or at $s = \infty$ in s domain.

Use initial value theorem to find the initial value of $F(s)$, here $f(0)$ indicates the initial value of $F(s)$ in time domain.

$$f(0) = \lim_{s \to \infty} sF(s)$$

Step 2 of 6

Therefore, initial value of $F(s)$ is,

$$\begin{aligned} f(0) &= \lim_{s \to \infty} sF(s) \\ &= \lim_{s \to \infty} s\frac{5s^2 + 3}{s^3 + 4s^2 + 6} \\ &= \lim_{s \to \infty} \frac{s(5s^2 + 3)}{s^3 + 4s^2 + 6} \\ &= \lim_{s \to \infty} \frac{s^3\left(5 + \frac{3}{s^2}\right)}{s^3\left(1 + \frac{4}{s} + \frac{6}{s^2}\right)} \end{aligned}$$

$$= \frac{\left(5 + \frac{3}{\infty^2}\right)}{\left(1 + \frac{4}{\infty} + \frac{6}{\infty^2}\right)}$$

$$= \frac{(5+0)}{(1+0+0)}$$

$$= 5$$

Therefore, the initial value of the function $F(s)$ is $\boxed{5}$.

Step 3 of 6

Final value theorem is applicable to the functions which are having all the poles in the left half of the s plane.

The poles of s $F(s)$ are located at $s=-4.3213$, $0.1607+j1.1673$, $0.1607-j1.1673$. So, all the poles are not located in left half of the s plane.

Hence, the final value theorem is not valid.

Therefore, final value of $F(s)$ doesn’t exist.

Step 4 of 6

(b)

The function $F(s)$ is,

$$F(s)=\frac{s^2-2s+1}{4(s-2)(s^2+2s+4)}$$

The initial value of any function is the value obtained at time $t=0$ in time domain or at $s=\infty$ in s domain.

Use initial value theorem to find the initial value of $F(s)$, here $f(0)$ indicates the initial value of $F(s)$ in time domain.

$$f(0)=\lim_{s\to\infty} sF(s)$$

Therefore, initial value of $F(s)$ is,

$$\begin{aligned} f(0) &= \lim_{s\to\infty} sF(s) \\ &= \lim_{s\to\infty} s\frac{s^2-2s+1}{4(s-2)(s^2+2s+4)} \\ &= \lim_{s\to\infty} \frac{s(s^2-2s+1)}{4(s-2)(s^2+2s+4)} \end{aligned}$$

$$= \lim_{s\to\infty} \frac{s^3\left(1-\frac{2}{s}+\frac{1}{s^2}\right)}{4s^3\left(1-\frac{2}{s}\right)\left(1+2\frac{1}{s}+\frac{4}{s^2}\right)}$$

$$= \lim_{s\to\infty} \frac{\left(1-\frac{2}{s}+\frac{1}{s^2}\right)}{4\left(1-\frac{2}{s}\right)\left(1+\frac{2}{s}+\frac{4}{s^2}\right)}$$

$$= \frac{\left(1-\frac{2}{\infty}+\frac{1}{\infty^2}\right)}{4\left(1-\frac{2}{\infty}\right)\left(1+\frac{2}{\infty}+\frac{4}{\infty^2}\right)}$$

Step 5 of 6

Further simplification gives,

$$\begin{aligned} f(0) &= \frac{(1-0+0)}{4(1-0)(1+0+0)} \\ &= \frac{1}{4} \end{aligned}$$

Therefore, the initial value of the function $F(s)$ is $\boxed{\frac{1}{4}}$.

Final value theorem is applicable to the functions which are having all the poles in the left half of the s plane.

The poles of s $F(s)$ are located at $s = 2, -2, -2$. So, all the poles are not located in left half of the s plane.

Hence, the final value theorem is not valid.

Therefore, final value of $F(s)$ doesn’t exist.